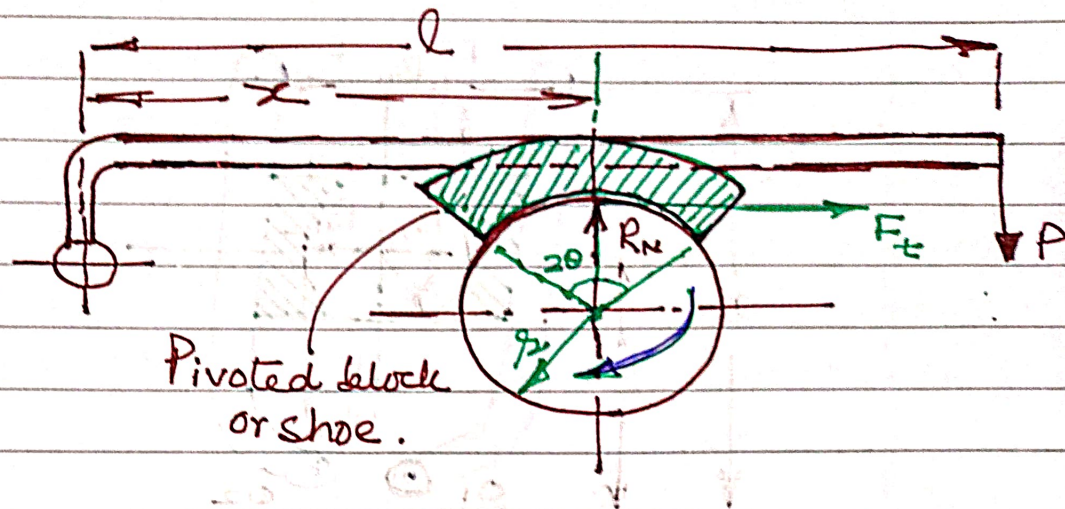


②

Pivoted Block or Shoe Brake:

Not Recd
Date _____
Page _____

①



When the angle of contact is greater than 60° , then the unit pressure normal to the surface of contact is less at the ends than at the centre. In such cases, the block or shoe is pivoted to the lever, instead of being rigidly fixed to the lever. which gives uniform wear to the brake lining in the direction of the applied force.

when $2\theta > 60^\circ$

$$\therefore \text{Braking Torque } T_B = F_t \times r = M' R_N \times r$$

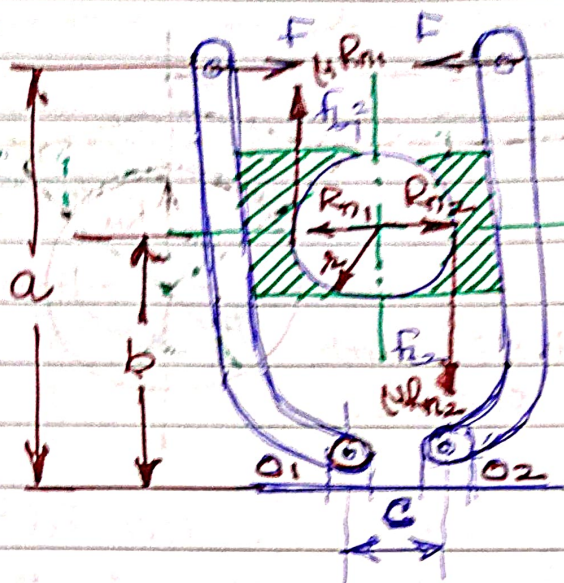
where $M' = \text{Equivalent Coefficient of friction} = \frac{4\mu \sin \theta}{2\theta + \sin 2\theta}$

& $\mu = \text{Actual Coefficient of friction.}$

⑧

Double Block or Shoe Brake

②



When a single block brake is applied to a rolling wheel, an additional load is thrown on the shaft bearings due to normal force (R_n). This produces bending of the shaft. In order to overcome this drawback, a double or shoe brake is used.

It consists of two ~~the~~ block brake applied at the opposite ends of a diameter of the wheel which eliminates or reduces the unbalanced force on the shaft.

Braking Torque

$$T_B = (F_{f1} + F_{f2}) r$$

Simple Band Brake

It consists of a flexible band of leather, one or more ropes, or a steel lined with friction material, which embraces a part of the circumference of the drum.

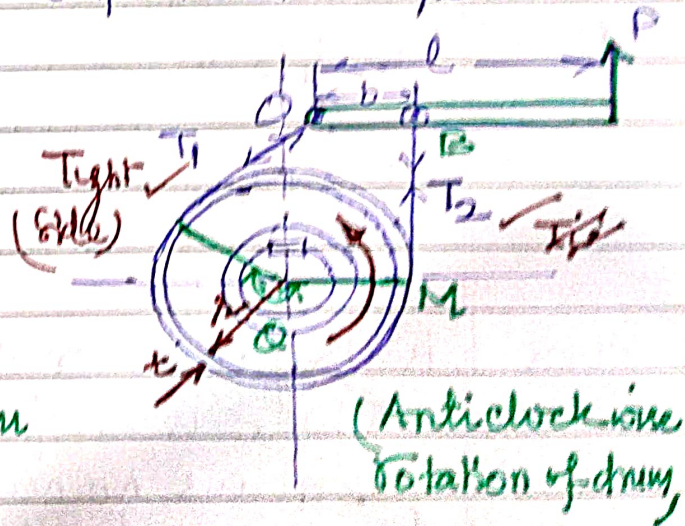
In Simple band brake, one end of the band

Hand: $T_2 \rightarrow \dots \rightarrow n \quad \text{Al} \quad n \quad \text{Glack} \quad n \quad n \quad n \quad n$

$\mu \rightarrow$ Coeff. of friction

$t \rightarrow$ thickness of band

Diagram illustrating the process of a gas molecule colliding with a piston and exerting pressure. A piston is shown moving up and down, with a green arrow labeled 'P' indicating the force exerted by the gas molecules on the piston.

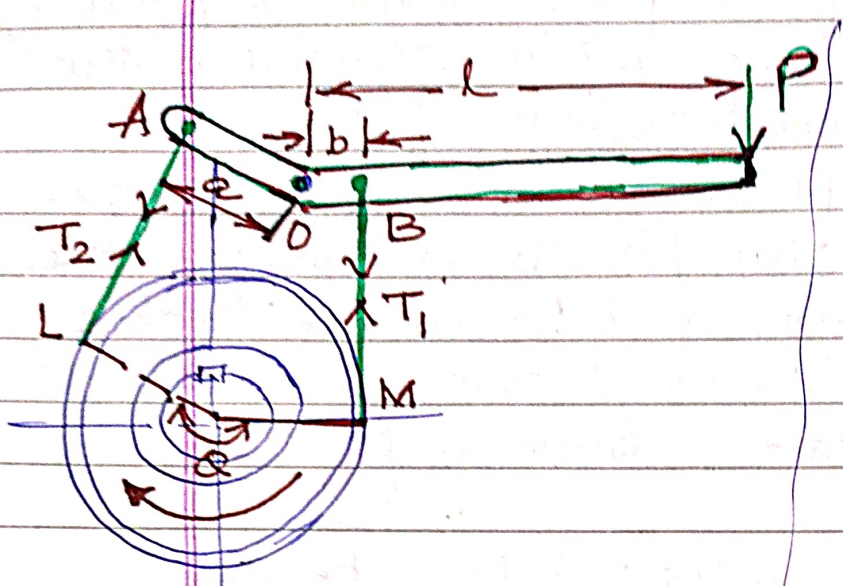


✓ $T_1 = e^{140}$
 T_2

$$= (T_1 - T_2) \text{ re } \rightarrow \text{ (Considering thickness of drum)}$$

$p.l = T_2 b \rightarrow ("Anti")$

Differential Band Brake :-



(Clockwise rotation of drum)

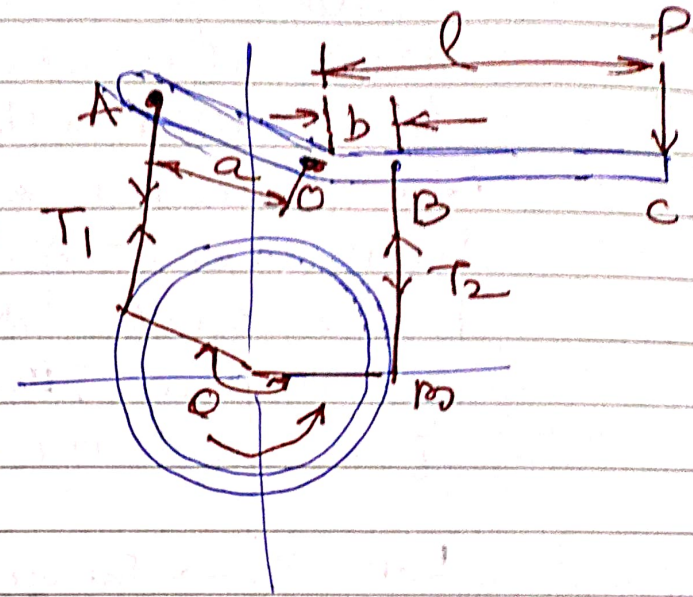
$$P \cdot l + T_1 b = T_2 a$$

$$P = \frac{T_2 a - T_1 b}{l}$$

$$\text{As } T_2 a \leq T_1 b$$

$$\text{or } \frac{T_1}{T_2} \geq a/b$$

When P is negative or zero, then Brake is self locking, means no need of force to apply brake.



(Anticlockwise rotation of drum)

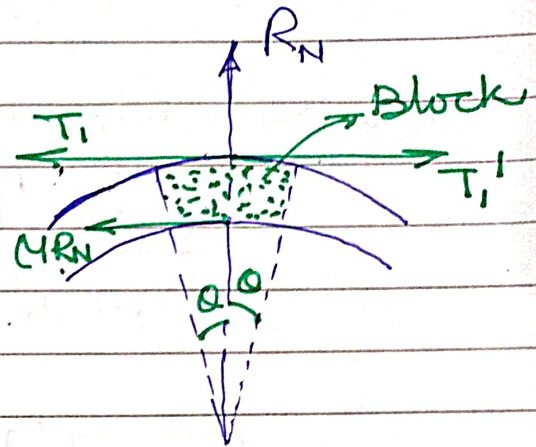
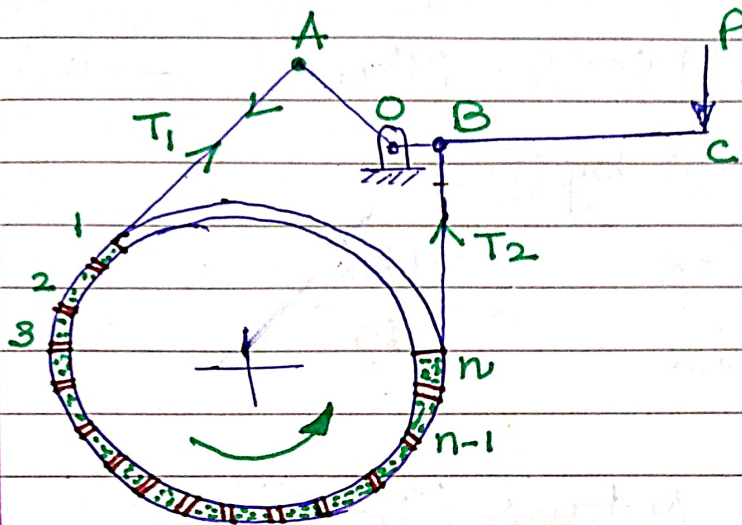
$$P \cdot l + T_2 b = T_1 a$$

$$P = \frac{T_1 a - T_2 b}{l}$$

$$T_2 b \geq T_1 a$$

$$\frac{T_2}{T_1} \geq a/b$$

Band or Block Brake →



In Such type of brake may be lined with blocks of wood or other material.

The friction between blocks and the drum provides braking action.

Let n number of blocks, each subtending an angle 2θ at the Centre and the drum rotates in Counterclockwise direction.

T_1 = Tension in tight side

T_2 = " " " slack "

μ = Coeff. of friction b/w blocks and drum.

T_1' = Tension in band b/w 1 & 2 block.

$T_2', T_3' = \dots$ " " " 2 & 3 and 3 & 4 block etc.

Consider first block, which is in Equilibrium under the following forces.

- (1) T_1 (Tension in tight side).
- (2) Tension in band b/w 1 & 2 block (T_1')
- (3) Normal reaction R_N
- (4) Frictional force μR_N .

Resolving forces radially,

$$(T_1 + T_1') \sin \theta = R_N. \quad \text{--- (1)}$$

Resolving the forces tangentially

$$(T_1 - T_1') \cos \theta = MR_n \quad \text{--- (2)}$$

Dividing Eqⁿ (2) by (1),

$$\frac{(T_1 - T_1') \cos \theta}{(T_1 + T_1') \sin \theta} = \frac{MR_n}{R_n}$$

$$\therefore (T_1 - T_1') = M \tan \theta (T_1 + T_1')$$

$$\frac{T_1}{T_1'} = \frac{1 + M \tan \theta}{1 - M \tan \theta}$$

Similarly,

$$\frac{T_1'}{T_2'} = \frac{T_2'}{T_3'} = \frac{T_3'}{T_4'} = \dots = \frac{T_{n-1}}{T_2} = \frac{1 + M \tan \theta}{1 - M \tan \theta}$$

$$\therefore \frac{T_1}{T_2} = \frac{T_1}{T_1'} \times \frac{T_1'}{T_2'} \times \frac{T_2'}{T_3'} \times \dots \times \frac{T_{n-1}}{T_2} = \left(\frac{1 + M \tan \theta}{1 - M \tan \theta} \right)^n$$

Braking torque on drum of effective radius.

$$\checkmark \quad T_B = (T_1 - T_2) R_e$$

$$\checkmark \quad \& \quad T_B = (T_1 - T_2) R \quad (\text{Neglecting thickness of band.})$$

————— x ————— x —————